Timestamp Health in Multi-Sensor Driving Data: Measuring Jitter, Drift and Offset on KITTI, and What They Imply for ROS 2 Recording Pipelines

Project 1 of a three-part study of silent failure modes in robot data pipelines

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Draft v0.1 · Preprint (not peer reviewed)

Abstract

Multi-sensor perception stacks associate camera frames, LiDAR sweeps and inertial data by timestamp. Each sensor stamps data from its own clock, through its own software path, and when a stamp is wrong nothing fails: the data is simply mis-associated. This report measures three timing faults, offset, drift and jitter, on three KITTI raw drives (16–33 s) and on synthetic faults with known ground truth. On real data the camera and LiDAR stamps are tight (robust interval jitter 0.19 ms and 0.08 ms), but the OXTS inertial stream contains three backward timestamp steps, ten repeated stamps and two gaps. A nearest-timestamp pairing estimator recovers injected drift exactly when stamps are noise-free, but with 2–10 % dropped frames its error standard deviation grows from about 8 ppm to 219–446 ppm unless mis-paired samples are gated; gating restores 8–10 ppm. In simulation, resolving a 2 ppm drift needs roughly 300 s of data, far longer than the drives analysed, so the health report must be able to say "not resolvable at this session length". Each uncorrected millisecond costs 0.45 cm of position and 0.009° of attitude on this drive. A gyro-versus-visual yaw estimator reports a consistent apparent camera-to-IMU lag of 64–78 ms across three drives, which is not what the dataset's hardware synchronisation would lead one to expect; it changed sign when the camera intrinsics were corrected, so we treat it as an unverified estimator output. We discuss what these findings suggest for ROS 2 recording and fusion pipelines and outline a measurement protocol for follow-up work.

Keywords: timestamp synchronisation; clock drift; jitter; temporal calibration; sensor fusion; data validation; KITTI; ROS 2; MCAP; dataset quality

1 Introduction

A driving or mobile-robot dataset is only useful for 3D perception if the sensors in it are correctly related in time as well as in space. To fuse a LiDAR sweep with a camera frame, a stack must answer a simple question: where was the vehicle at this camera timestamp? The answer is only as good as the timestamps. If the camera stamp is off by 29 ms, then at 30 m/s the vehicle has moved 87 cm, and a point cloud and an image that the model believes show the same moment do not. A practitioner account of robotics data infrastructure gives exactly this example, attributing it to a 2 ppm oscillator drift accumulated over a four-hour session [1]. The arithmetic is easy to confirm ($2\,\text{ppm} \times 14\,400\,\text{s} = 28.8\,\text{ms}$; $28.8\,\text{ms} \times 30\,\text{m/s} = 86\,\text{cm}$), and what makes the problem serious is not its size per frame but its persistence: it applies, slightly and consistently, to every frame from the affected sensor.

What makes timing faults hard to deal with is that they are silent. A stamp that is 29 ms late is still a valid integer. A repeated stamp, a stamp that steps backwards, or a dropped frame passes

through most ingestion code without an exception. The model trains, the loss decreases, and the damage appears later as an unexplained regression. This is the pattern that Sambasivan et al. call a data cascade: upstream data problems that compound through the development process, surface downstream, and come with no clear indicators, tools or metrics for detecting them [2].

Temporal calibration for camera–IMU systems has a substantial literature [3, 4, 5, 6], and data validation for machine learning has another [7, 8]. They meet only weakly. Calibration methods assume a clean, monotone stream and estimate a time offset inside an estimator; data-validation systems check schemas and distributions but have little to say about whether a timestamp column is trustworthy. This report asks a narrower, practical question: given only the raw timestamp streams of a recorded session, how healthy are they, how large a fault can we detect, and what does an undetected fault cost?

Contributions. Using three KITTI raw drives [9] and a fault-injection harness with known ground truth, we:

- audit the real timestamp streams of camera, LiDAR and IMU and document the anomalies present in an unmodified, widely used dataset (Section 4.1);
- show that nearest-timestamp pairing, the default for cross-sensor association, fails badly under dropped frames, and that a simple pairing gate removes the failure (Section 4.3);
- quantify how long a session must be before a given drift is statistically distinguishable from zero (Section 4.4);
- measure the cost of an uncorrected offset on the pose stream, per millisecond (Section 4.5);
- report an attempt at gyro-versus-visual temporal calibration on short drives, including where the estimator is weak, and what it does and does not establish (Section 4.6);
- translate the findings into reasoned, explicitly untested implications for ROS 2 recording and fusion pipelines (Section 5).

This is Project 1 of three. Project 2 covers silent data corruption and dataset hygiene (checksums, duplicates, split leakage, label conventions); Project 3 covers provenance, calibration tracking and ROS 2 latency. The projects are written to stand alone.

2 Background and related work

2.1 Where timestamps come from, and how they go wrong

A timestamp is a reading of some clock at some point in the data path. In a typical robot, the camera may stamp frames from its own oscillator, the LiDAR from another, the IMU from a third, and the host operating system applies a fourth stamp at receipt. GPS with a pulse-per-second signal provides an absolute reference, and network protocols such as NTP [10] and IEEE 1588 PTP [11] discipline host clocks, but they do not discipline a free-running oscillator inside a sensor. Three faults follow. A constant *offset* arises from trigger and transmission delay or from different stamping conventions. *Drift* arises because oscillators have a frequency error: 1 ppm is 1 µs per second, or 3.6 ms per hour. *Jitter* is the random scatter of a stamp around its true time, from scheduling, buffering and transport. Practitioners quote 1–5 ppm for camera oscillators [1]; we did not measure oscillators and use these figures only to set the range of injected faults. Table 1 shows how drift accumulates and what it costs at highway speed.

2.2 Temporal calibration in visual-inertial estimation

Kelly and Sukhatme gave an early general framework for temporal calibration across proprioceptive and exteroceptive sensors [3]. Li and Mourikis made the camera–IMU time offset an

Table 1: Offset accumulated by a constant clock drift, and the resulting position error for a vehicle at 30 m/s (position error = speed × offset). Computed arithmetic, not a measurement.

Drift	Accumulated after 1 h	Position error at 30 m/s	Accumulated after 4 h	Position error at 30 m/s
1 ppm	3.6 ms	11 cm	14.4 ms	43 cm
2 ppm	7.2 ms	22 cm	28.8 ms	86 cm
5 ppm	18.0 ms	54 cm	72.0 ms	216 cm

additional state in an EKF-based estimator and analysed when it is identifiable, showing that it can be estimated online together with the other quantities of interest [4]. Furgale, Rehder and Siegwart estimated temporal offsets jointly with spatial transformations using continuous-time batch estimation, an approach associated with the Kalibr calibration toolbox [5]. Qin and Shen extended the idea to monocular visual-inertial SLAM, estimating a constant offset online within a feature-based optimisation, motivated by the triggering and transmission delays that cause temporal misalignment between sensors, and released it inside VINS-Mono [6].

These works share three features that matter here. They treat the offset as a constant or slowly varying parameter; they estimate it inside a full estimator that already has a good motion model; and they presuppose well-formed streams. None is designed to tell a data engineer, before training, whether a recorded session contains repeated stamps, backward steps, gaps or drift, or how many seconds of data are needed before a drift can be seen at all. This report is about that earlier question, and its camera–IMU experiment (Section 4.6) is deliberately simpler than the estimators above, and weaker.

2.3 Data quality as an engineering discipline

Sculley et al. argued that machine-learning systems accumulate hidden technical debt, including in data dependencies [12]. Breck et al. described a schema-based data-validation system that catches anomalies before training [7], and Schelter et al. proposed declarative "unit tests for data" at scale [8]. Work on benchmark hygiene shows the cost of undetected faults: label errors in widely used test sets [13], near-duplicate images shared between training and test splits [14], duplicates in language-model corpora [15], and leakage as a driver of irreproducible results across scientific fields [16]. Timestamps are rarely named in this literature, although they are the key that joins every sensor stream. We treat timestamp health as a data-validation problem and, because faults can be injected with known magnitude, as a convenient first case in which a validator's sensitivity can be measured rather than asserted.

2.4 The middleware layer

ROS 2 is the dominant robotics middleware [17]. Messages carry a header stamp assigned by the publisher, are delivered by a DDS transport, and are processed by callbacks that the executor schedules. Casini et al. show that end-to-end latency of ROS 2 processing chains depends on the executor's callback scheduling and on operating-system scheduling beneath it [18]. This matters for timing data because any stamp applied after those layers, for example at receipt by a recorder, includes their variability. Recordings are commonly stored in MCAP [19], whose message records carry both a log time and a publish time alongside the payload, so more than one clock is available per message. Section 5 returns to what this allows.

3 Data and methods

3.1 Data

We use KITTI raw [9] drives `2011_09_26_drive_0005` (160 camera frames, 16.4 s), `0011` (239 frames, 24.6 s) and `0014` (320 frames, 33.0 s). Streams are the left grey camera (`image_00`), the OXTS GPS/IMU unit, and the Velodyne LiDAR. The KITTI platform triggers its cameras from the spinning LiDAR so that the streams are nominally synchronised [9], which makes the dataset a best case for synchronisation: real-world recordings without such a trigger should be worse. Drive 0005 is used for all LiDAR experiments; drives 0011 and 0014 are used only to replicate the camera–IMU offset estimate. All stamps are converted to integer nanoseconds. No KITTI data is redistributed with this report (Section 7).

3.2 Stamp hygiene

For each stream we compute inter-sample intervals and flag non-positive intervals, intervals above a gap threshold (15 ms for the 10 ms IMU stream), and stamps that appear more than once, comparing the payloads of repeated stamps. Jitter is reported as the robust standard deviation of the interval (1.4826× median absolute deviation) so that a few outliers do not dominate.

3.3 Pairing, offset and drift estimation

For each camera stamp we find the nearest stamp in the other stream and record the signed difference. Pairs outside the overlap of the two streams are removed. Offset is the median difference; drift is the slope of difference against time, estimated by ordinary least squares with a 95 % interval from the residual variance and, because the residuals are autocorrelated, a block bootstrap (blocks of 10). The Theil–Sen estimator [20] is used as a robust comparison. To handle dropped frames, pairs whose difference is more than 30 ms from the median are rejected as mis-pairs; the true scatter is about 1 ms whereas a mis-pair is a whole frame period (about 103 ms) away.

3.4 Fault injection

A single function perturbs one stream's timestamps with a known offset, linear drift in ppm, Gaussian jitter and a dropped-frame fraction (isolated or in bursts of three), with fixed seeds. The perturbed stream is then passed through the same estimator, and the recovered drift is compared with the injected value. Monte-Carlo runs use 100–200 seeds. Detectability versus session length is studied with a separate synthetic generator calibrated to the observed camera and LiDAR interval statistics (camera jitter 0.8 ms with occasional one-sided dips, LiDAR jitter 0.1 ms).

3.5 Timestamp-health verdict and score

The health report combines the dropped-frame fraction, robust jitter and drift into a score

$$S = 100 \cdot \exp(-0.25\,p), \tag{1}$$

where p sums dropped-frame percentage (1 % = 1 unit), jitter (1 ms = 1 unit) and, only if the drift is statistically detectable and its accumulated position cost exceeds a tolerance (5 cm), the ratio of that cost to the tolerance. Drift receives a separate verdict: OK if even the 3σ worst case stays within tolerance, EXCEEDS if the detected drift exceeds it, and UNRESOLVED

otherwise. These weights are a design choice and have not been calibrated against downstream model accuracy; the verdict is the more defensible output.

3.6 Cost of an uncorrected offset

We shift camera stamps by a known amount and compare the OXTS-derived pose (position from latitude/longitude on a local Mercator projection; attitude by spherical interpolation of roll, pitch and yaw) at the shifted and original stamps. The RMS error over camera frames is compared with the analytic value $v \cdot \Delta t$ using the drive's RMS speed (4.63 m/s). This measures the geometry error that a mis-stamped frame inherits; it is not a trained model's accuracy.

3.7 Camera–IMU temporal calibration from gyro and visual yaw

For each consecutive camera pair we detect ORB features [21] in the upper 80 % of the image (to exclude the car hood), match them, and estimate the essential matrix with RANSAC [22, 23] on points undistorted with the KITTI calibration (K, D of camera 00). The recovered relative rotation gives a visual yaw increment per camera interval. The gyro yaw rate about the body-up axis is integrated over the same interval shifted by a candidate $\tau \in [-120, 120]$ ms in 1 ms steps. Three cost functions are compared: Pearson correlation, RMS residual with the gain fixed at 1 (justified once calibrated intrinsics give a fitted gain of 0.99), and median absolute residual. The minimum is refined by a parabola fit. Pairs are kept only if they have at least 100 inliers, are not within 0.25 s of an IMU gap, and the recovered translation direction is within 12° of the optical axis (a forward-driving car should translate roughly along it; pairs failing this have an unreliable decomposition, Section 4.6). Uncertainty comes from a bootstrap over pairs. This follows the spirit of cross-correlating angular rates, but it is far simpler than the continuous-time or filter-based estimators in Section 2.2.

3.8 Fusion proxy

To relate stamp error to a fused estimate, we run a three-state yaw filter (yaw, gyro bias, yaw at the previous camera frame) that propagates yaw with the gyro and updates it with the visual yaw increment (measurement noise 0.2°). OXTS yaw is used as truth. A gyro-only run provides the baseline. This is a proxy only, because the truth and the gyro come from the same IMU.

4 Results

4.1 Real timestamp streams are not as clean as the camera suggests

Figure 1 shows the inter-sample intervals. The LiDAR is smooth, with its period wandering between 102.7 and 103.7 ms (std 0.12 ms). The camera period sits near 103.3 ms but with structured excursions to 100.2 and 106.6 ms (std 0.91 ms), so a few frames are stamped about 3 ms early or late. The IMU stream, nominally 10 ms, contains anomalies near sample 1030–1060 (Table 2).

The OXTS anomalies are of three kinds. There are three backward steps (−49.99, −49.84, −50.19 ms) and 10 stamps that occur more than once, each time with an identical payload, so that 1648 raw samples reduce to 1633 unique stamps. The pattern, a gap followed by a step back and a re-run of earlier stamps, is consistent with a buffered burst being delivered twice, but we did not investigate the cause. A monotonicity check at ingestion would have caught it. Streams also start at different times: the IMU begins 43.0 ms and the LiDAR 188.7 ms after the camera.

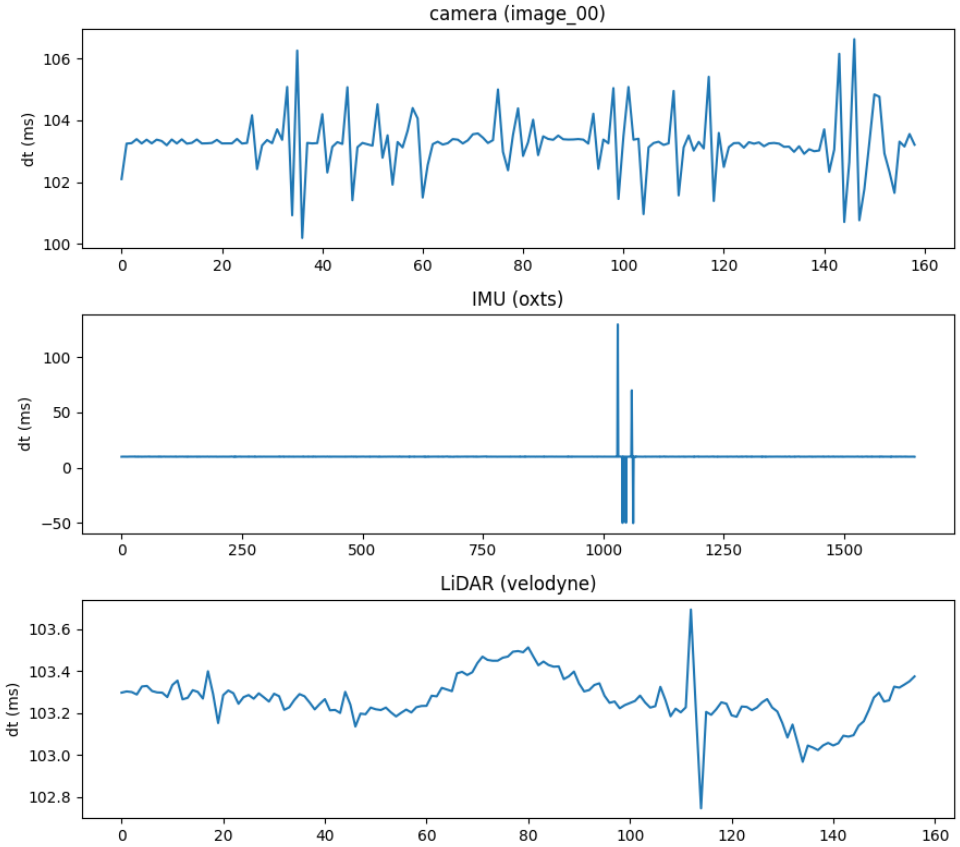


Figure 1: Inter-sample intervals on drive 0005. Top: camera (`image_00`). Middle: OXTS IMU, with a 130 ms gap, a 70 ms gap and four backward steps of about −50 ms. Bottom: Velodyne LiDAR. Note the different vertical scales.

4.2 Offsets, and a sampling artefact that looks like jitter

The camera-to-LiDAR offset is nearly constant at −17.0 ms after trimming (std 0.78 ms; Figure 2, top), which we read as a difference in stamping convention rather than a fault. The camera-to-IMU offset has mean 0.01 ms and std 2.94 ms, but this scatter is quantisation: matching each camera stamp to the nearest of 10 ms IMU samples produces an error uniformly distributed over ±5 ms, whose standard deviation is $10/\sqrt{12} = 2.89$ ms. The saw-tooth in Figure 2 (bottom) is therefore not evidence of a jittery IMU clock. Any health score that reports "jitter" from nearest-neighbour offsets against a slower stream will overstate it.

4.3 Drift estimation, injected faults and dropped frames

Over the 16 s overlap the camera-to-LiDAR slope is −33 ppm by least squares (±26 ppm, 95 %; block-bootstrap interval $[-74, 22]$ ppm), so zero drift cannot be rejected. The residual std is 0.767 ms for a linear and 0.766 ms for a quadratic fit, so there is no evidence of curvature. Theil–Sen gives −5 ppm $[-10, -1]$ on the same data. The two estimators disagree because the offsets are not symmetric noise: Figure 3 shows one-sided dips of 1–3 ms (the camera-stamp excursions of Section 4.1), which pull a least-squares line and are only partly rejected by a rank-based one. With injected Gaussian jitter of 0.5 ms the least-squares estimator is unbiased (mean error 0.2 ppm, std 8.2 ppm over 200 seeds), whereas Theil–Sen shows a systematic −14 ppm shift, so the gated least-squares estimator is used throughout.

With noise-free injected stamps the Theil–Sen estimator recovers 0, 20 and 200 ppm to within

Table 2: Interval statistics for drive 0005. Cleaning removed repeated stamps and sorted the stream; two gaps remain (129.8 ms, about 12 samples missing; 40.0 ms, about 3 missing).

Stream	n	mean dt (ms)	std dt (ms)	min / max dt (ms)	robust jitter (ms)
Camera	160	103.260	0.910	100.19 / 106.63	0.19
OXTS IMU (raw)	1648	10.000	4.179	−50.19 / 129.81	—
OXTS IMU (cleaned)	1633	—	—	9.77 / 129.81	—
LiDAR	158	103.267	0.116	102.75 / 103.69	0.08

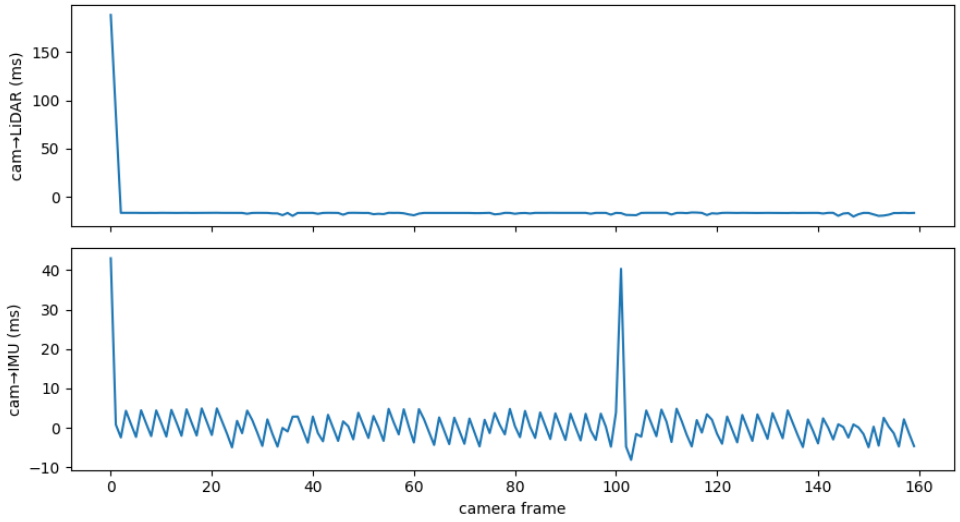


Figure 2: Signed nearest-neighbour offset from each camera stamp to LiDAR (top) and IMU (bottom), drive 0005. The first sample in each panel is a start-up artefact (the other stream begins later). The bottom spike near frame 100 coincides in time with the IMU gap. The saw-tooth is quantisation by the 10 ms IMU grid.

0.1 ppm, so any error under jitter comes from the data, not the estimator's arithmetic. Dropped frames are the harder case (Table 3, Figure 4a). If a LiDAR frame is missing, the nearest stamp to a camera frame is its neighbour, about 103 ms away. These mis-pairs are far from the true scatter and dominate the fit: with 2 % of frames dropped the error std rises from 8 ppm to 219 ppm, and to 446 ppm at 10 %. The 30 ms gate removes the mis-pairs at the cost of 2–10 % of the pairs and returns the error to 8–10 ppm.

4.4 How long a session is needed to see a drift

Because the standard error of a regression slope falls rapidly with the time span, short sessions cannot resolve small drifts, however clean the stamps. In simulation (Table 4, Figure 4b), the slope noise is 17.2 ppm at 16 s, 2.6 ppm at 60 s and 0.23 ppm at 300 s. Requiring three standard deviations to lie below 2 ppm needs roughly 300 s. All three real drives are well under this, and the health report should therefore state a drift verdict of "UNRESOLVED (session too short)" rather than reporting a small number as if it were measured. The simulation uses assumed jitter statistics and a constant drift; real drift can vary with temperature, so this is a lower bound on required session length.

4.5 What an uncorrected millisecond costs

Shifting stamps on the real OXTS pose stream gives position errors that match the analytic $v \cdot \Delta t$ to within 2–3 % (Table 5), and an attitude error of 0.0094° per millisecond. The cost is

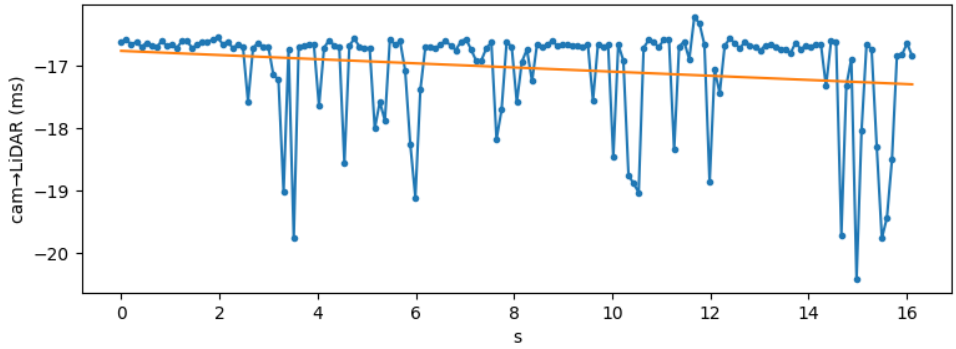


Figure 3: Camera-to-LiDAR offset against time on drive 0005 with the least-squares fit (−33 ppm). The one-sided dips are camera-stamp excursions; they make the offset distribution skewed.

Table 3: Error in recovered drift (estimate minus baseline) for the naive nearest-stamp pairing versus the gated pairing, with 0.5 ms injected LiDAR jitter and 0 ppm injected drift, 100 seeds per row. "Burst" is the number of consecutive frames dropped per event.

Dropped	Burst	Naive: mean / std (ppm)	Gated: mean / std (ppm)	Pairs kept (gated)
0 %	1	0.4 / 8.0	0.4 / 8.0	1.00
2 %	1	11.5 / 219.0	0.5 / 8.3	0.98
2 %	3	1.3 / 264.5	0.8 / 8.4	0.98
5 %	1	1.2 / 347.8	0.5 / 8.5	0.95
5 %	3	67.0 / 380.3	1.5 / 9.4	0.95
10 %	1	28.7 / 446.3	−0.1 / 9.8	0.90
10 %	3	58.3 / 518.2	1.5 / 10.2	0.90

linear and small per millisecond: 10 ms is 4.5 cm on a drive with an RMS speed of 4.6 m/s. At 30 m/s the same offset is 30 cm (arithmetic from the same relation). A drift that accumulates to 10 ms therefore has the same geometric effect as a constant 10 ms offset, but it grows throughout the session.

4.6 Camera–IMU offset from gyro and visual yaw: what it does and does not show

This was the hardest part of the study and the least conclusive, so we report its course. With approximate intrinsics and raw pixels (Figure 5), visual yaw correlates with gyro yaw ($r = 0.79$ at zero shift), and the correlation peaks at −85 ms. But the correlation varies by only 0.012 over the whole ±120 ms search window, the bootstrap interval is $[-107, 30]$ ms with 36 % of resamples failing to produce a peak, and simulations at the observed visual noise (0.8°) find the estimate within ±10 ms in only 6–7 % of trials with 145 pairs, and in 55–58 % even with 9300 pairs (about 16 minutes of data). The information is simply weak: gyro yaw per camera interval is smooth, so shifting the window by tens of milliseconds changes little.

Two changes improved the signal. Replacing approximate intrinsics with the calibrated K and D reduced the yaw residual (MAD-sigma 0.46° → 0.17°) and moved the fitted visual-to-gyro gain from 1.21 to 0.99. The residuals then concentrated in pairs whose recovered translation was far off the optical axis: in the top quartile of that angle (12°–88°) the residual std is 2.0°, against 0.10–0.15° in the other three quartiles. Gating on that angle (keeping 108 of 145 pairs) and switching to a fixed-gain RMS cost gives a minimum at +78.1 ms with bootstrap interval $[46, 102]$ ms (Figure 6, Table 7). The median-absolute-residual cost is too noisy to be useful (interval width 121 ms).

Table 4: Simulated detectability of a 2 ppm drift against session length (200 seeds per column; camera jitter 0.8 ms with one-sided dips, LiDAR jitter 0.1 ms). Estimator bias is below 0.15 ppm in every column.

Session length	16 s	60 s	120 s	300 s	1200 s	3600 s
Slope noise, std (ppm)	17.18	2.61	0.92	0.23	0.03	0.01
3σ (ppm)	51.55	7.84	2.76	0.69	0.09	0.02
2 ppm detectable?	no	no	no	yes	yes	yes

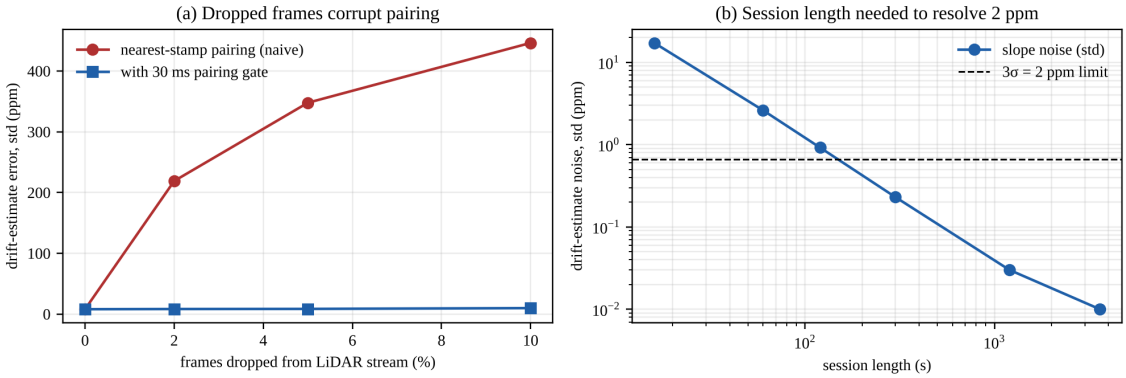


Figure 4: (a) Drift-estimate error against the fraction of dropped LiDAR frames, naive versus gated pairing (values from Table 3, isolated drops). (b) Simulated drift-estimate noise against session length; the dashed line marks the level at which 3σ equals 2 ppm.

The estimates agree across drives (64–78 ms, overlapping intervals, none containing zero). That rules out sampling noise as the explanation, but it does not establish that KITTI's camera stamps are 70 ms off. Three observations argue for caution. First, the nearest-sample comparison of Section 4.2 shows camera and IMU stamps coincide to within the quantisation of the 10 ms IMU grid, which is what the hardware trigger would lead one to expect. Second, the sign of the estimate flipped between the approximate-intrinsics run (−85 ms) and the calibrated run (+78 ms), so the estimator is sensitive to preprocessing. Third, a systematic bias common to all three drives, for example a mismatch in how an image interval is defined, an axis or sign convention in the camera-to-body mapping, or unmodelled rolling or exposure effects, would produce exactly this consistency. We therefore report a consistent apparent lag, not independently verified. A joint estimator that uses all three rotation axes and the KITTI IMU-to-camera extrinsics would be the natural next check.

The fusion proxy gives a consistent but weak picture (Table 8). With the estimated correction applied, the filter's RMS yaw error is 0.606°, 0.61 times that of gyro-only propagation (0.991°); with raw stamps it is 0.730° (0.74×). The error rises by about 0.003° per millisecond of residual stamp error on the positive side, so across ±20 ms of residual error it varies only between 0.56° and 0.65°. The filter error is lowest at $s = -20$ ms, a shift of about 58 ms ($s = -50$ ms, about 28 ms, is nearly as good), which is inside the bootstrap interval but below the point estimate. Differences below about 0.05° are not significant (single run, no confidence interval, truth and gyro share an IMU).

5 Implications for ROS 2 recording and fusion pipelines

Everything in this section is reasoning from the findings above and from the cited architecture work, not a measurement in ROS 2. Measuring it is the purpose of Project 3. We state it as hypotheses to be tested.

Table 5: Pose error caused by an uncorrected stamp offset on drive 0005 (RMS over camera frames; $v_{\mathrm{rms}} = 4.63\,\mathrm{m/s}$).

Offset (ms)	1	2	5	10	20	50
Position error RMS, measured (cm)	0.45	0.91	2.27	4.53	9.04	22.62
Position error, analytic $v \cdot \Delta t$ (cm)	0.46	0.93	2.31	4.63	9.26	23.15
Attitude error RMS (deg)	0.009	0.019	0.047	0.094	0.188	0.468

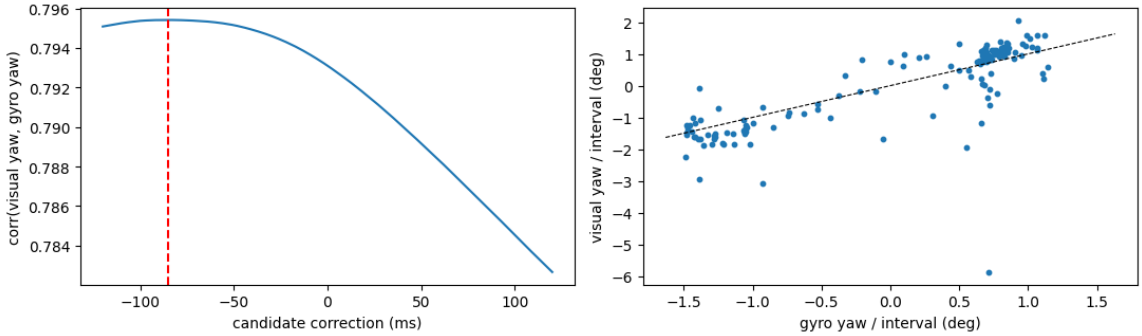


Figure 5: Left: correlation between visual yaw and gyro yaw increments against candidate correction, with approximate intrinsics (peak 0.795 at $-85\,\mathrm{ms}$, but the full range over $\pm 120\,\mathrm{ms}$ is only 0.783–0.795). Right: per-interval visual yaw against gyro yaw for 145 valid camera pairs; the dashed line is $y = x$.

5.1 Know which clock a stamp came from

A `header.stamp` set by a driver or sensor is close to the time of capture but lives on a clock that may drift relative to the host. A stamp applied by a node or recorder at receipt is on the host clock but folds in transport, executor and scheduling latency [18], which appears as jitter. The two failure modes look different in the data, as the injected-fault results suggest: a clock mismatch is a slope, scheduling is variance and a heavy tail. Because MCAP records a log time and a publish time beside the message [19], the difference between the header stamp and the log time, per topic, is a cheap health signal that needs no ground truth: its median is an offset, its slope is a drift, its scatter is jitter, and a gap in it is a drop.

5.2 Approximate-time association has the same failure as nearest-stamp pairing

Approximate-time synchronisers match messages from different topics by header stamp within a tolerance. Section 4.3 showed that nearest-stamp pairing breaks when a frame is missing, because the nearest neighbour is a different frame. We expect an approximate-time policy to be exposed to the same failure under drops (and to chains of mis-pairs when one topic is persistently late), but we did not test it. A tolerance gate and a counter of rejected pairs are inexpensive safeguards and make the failure visible.

5.3 Interpolation can hide jitter

A transform lookup that interpolates between samples will return plausible geometry for any query time inside its buffer. If jitter shifts the query stamp, the answer is smooth and wrong rather than an error. Timestamp validation therefore belongs before association, in the recording or ingest stage, where a failure can still be reported.

Table 6: Simulated precision of the Pearson-correlation lag estimator with 145 pairs: planted lag plus Gaussian visual noise, 300 repetitions per cell.

Visual noise (deg)	0.8	0.4	0.2	0.1
Std of estimate, true lag 0 ms (ms)	58.1	46.5	26.9	13.3
Std of estimate, true lag −85 ms (ms)	55.8	38.6	20.3	12.0
Failed estimates, true lag −85 ms	44 %	32 %	18 %	3 %

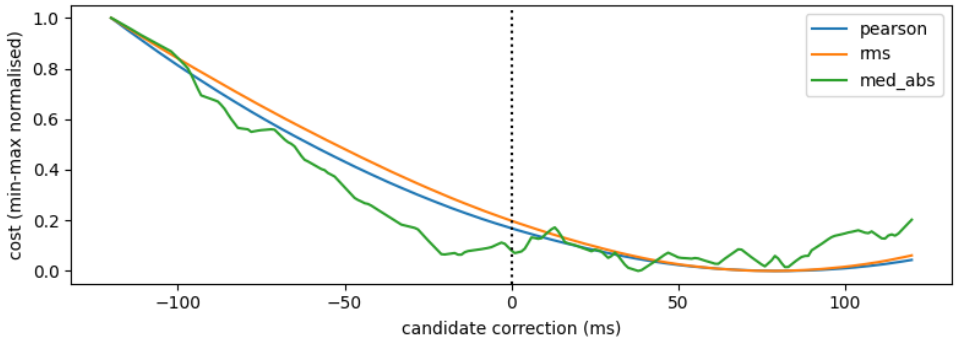


Figure 6: Normalised cost against candidate correction on drive 0005 after calibrated intrinsics and pair gating (108 pairs), for the three cost functions. Minima: Pearson +80.1 ms, RMS +78.1 ms, median absolute residual +38.4 ms. The curves are shallow: the RMS and Pearson costs stay within about 6 % of their range from roughly +55 ms to the +120 ms edge of the search window.

5.4 Design a health report that knows its own resolution

The results suggest four rules for a timestamp-health report in a recording pipeline: (i) check monotonicity, repeats and gaps first, since these were the only faults found in the real data without any injection; (ii) report jitter from intervals within a stream, not from nearest-neighbour offsets to a slower stream (Section 4.2); (iii) gate pairs before estimating offset or drift (Section 4.3); and (iv) give every drift estimate a session-length-dependent verdict (Section 4.4). Table 9 maps each signal to what it reveals and how it would be measured in ROS 2.

6 Limitations and threats to validity

- **Short, clean data.** The drives last 16–33 s and come from a hardware-triggered dataset. Real sessions last hours and have no such trigger, and drift that is invisible here may dominate there.
- **Synthetic faults.** Injected jitter is Gaussian and drift is linear. Real faults have structure: temperature-dependent drift, step changes after clock corrections, and bursty buffering. Recovery of injected faults shows the estimator is arithmetically sound, not that it is complete.
- **The camera–IMU result is unverified.** Consistency across three drives rules out noise but not systematic estimator bias; the sign changed with intrinsics; the cost curve is shallow; and the validation that injects a known shift into the camera stamps is circular because it only relabels the same grid. Only the noise simulations and the cross-drive replication carry information.
- **Weak fusion proxy.** The filter experiment uses OXTS yaw as truth, the same IMU as the gyro, from a single run, and it measures geometry error, not a trained model's accuracy.
- **Score weights are uncalibrated.** The health score is a design choice. We recommend reporting the components and the drift verdict, not the single number.
- **No ROS 2 measurements.** Section 5 is reasoned, not measured.

Table 7: Gyro-versus-visual lag estimate (RMS cost, calibrated intrinsics, lateral-angle gate) on three KITTI drives. Positive means the visual rotation best matches gyro data integrated later than the camera stamps.

Drive	Frames	Duration (s)	Pairs used	Offset estimate (ms)	Bootstrap 95 % interval (ms)
0005	160	16.4	108	78.1	[48.5, 101.6]
0011	239	24.6	117	64.2	[9.9, 98.5]
0014	320	33.0	285	71.9	[55.9, 84.9]

Table 8: Three-state yaw filter on drive 0005 with camera stamps shifted by $e_0 + s$, where $e_0 = +78.1$ ms is the estimated correction. $s = -78.1$ means uncorrected stamps. Single run.

Residual stamp error s (ms)	-78.1 (raw)	-50	-20	-10	-5	0	$+5$	$+10$	$+20$	$+50$
RMS yaw error (deg)	0.730	0.579	0.560	0.589	0.601	0.606	0.624	0.642	0.648	0.741
Ratio to gyro-only (0.991°)	0.74	0.58	0.57	0.59	0.61	0.61	0.63	0.65	0.65	0.75

7 Reproducibility and data availability

All results come from a single Python notebook run on CPU (Kaggle), using NumPy, SciPy, pandas, OpenCV and matplotlib; random seeds are fixed for injection and bootstrap. The notebook and figure files are available at [repository URL to be added] and archived at [Zenodo DOI to be added]. KITTI raw data are not redistributed; they can be obtained from the KITTI website under its own licence (CC BY-NC-SA 3.0), and the drives used are `2011_09_26_drive_0005`, `0011` and `0014`. Figures 1–3, 5 and 6 are taken directly from notebook output; Figure 4 is plotted from the values in Tables 3 and 4.

8 Conclusion

Timestamp faults are a data-quality problem with a measurable footprint. In a hardware-synchronised dataset the faults found without any injection were stream-integrity faults, backward steps, repeated stamps and gaps in the IMU, and a naive pairing step is the most fragile link: a few percent of dropped frames inflate its drift error by a factor of roughly 27–56 unless pairs are gated. A drift of a few ppm cannot be seen in less than several minutes of data, so the honest output of a health check on a short session is a verdict that states its own resolution. Each uncorrected millisecond costs 0.45 cm at 4.6 m/s, roughly 3 cm at highway speed, on every frame. The camera–IMU result is a cautionary example: a consistent, apparently significant offset that we cannot yet distinguish from a systematic estimator bias.

Next steps are a joint three-axis estimator with known extrinsics, validation on a dataset with independent ground-truth timing, and the ROS 2 measurements described in Section 5. Project 2 turns from time to integrity: checksums after write, near-duplicate frames, split leakage and label-schema compliance. Project 3 covers provenance and the middleware layer.

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Table 9: Timestamp-health signals, what they reveal, what this study measured and the analogue to be tested in ROS 2.

Signal	Reveals	Measured here	ROS 2 / MCAP analogue (to be tested)
Non-positive or repeated stamps	Replayed or buffered data	OXTS: 3 backward steps, 10 repeats	Per-topic monotonicity of header stamp and log time
Interval gaps	Dropped frames	OXTS: 2 gaps	Gap rate per topic, against nominal rate
Interval jitter (robust)	Stamping noise	Camera 0.19 ms, LiDAR 0.08 ms	Std of (log time − header stamp) per topic
Median pair offset	Constant offset	Cam–LiDAR −17 ms	Median of header-stamp differences between topics
Gated slope of offset	Clock drift	-33 ± 26 ppm (unresolved)	Slope of (log time − header stamp) over the session
Rejected-pair fraction	Mis-association	0–10 % in injections	Rejected fraction in an approximate-time policy

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